

The Strategy of Protectionism: Game Theory in International Trade

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Abstract

I apply concepts from game theory to analyze aspects of the 2025 Trade War. In the baseline game, the United States and China face the prisoner's dilemma and must choose to whether or not to levy tariffs. The Nash Equilibrium for the baseline game is $(Tariff, Tariff)$. In the first extension of the game, an indefinite repeated prisoner's dilemma where both players employ grim trigger is considered. I use discount rates to explain why the United States defected from the long trend of trade liberalization and why some foreign nations have not yet retaliated. The United States' discount rate has fallen below the threshold needed to sustain free trade, while the discount rate of countries who have not yet retaliated remains high. In the second extension, I use an indefinite game with alternating offers to model trade deals made by the U.S. and an Adversary. The new lower discount rate of the United States lowers their share of the gains from trade, resulting in a weaker outcome.

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I Introduction

The 2025 Trade War began on February 1, 2025, when the Trump Administration levied tariffs on Canada, Mexico, and China (Bown, 2025). Later on, the United States implemented several more tariffs, including global tariffs on all goods, and reciprocal tariffs on dozens of countries. The size of the trade war is unprecedented in the modern global economy. At its peak, just the tariffs levied by the United States and China on each other were 145 percent and 125 percent respectively (Bown, 2025). The announcement of the tariffs drew retaliation from several foreign countries and simultaneously opened negotiations with others. How can we model this trade war using game theory?

II The Baseline Game

The baseline game can be modeled as the prisoner’s dilemma, where the United States and China can choose from two strategies, either *Tariff* (T) or *No Tariff* (NT). In terms of the classic prisoner’s dilemma, *Tariff* is equivalent to *Defect*, and *No Tariff* is equivalent to *Cooperate*. The normal form of the game is as follows:

United States	China
$u_1(T, NT) = 3.4$	$u_2(NT, T) = 1.5$
$u_1(T, T) = 1$	$u_2(T, T) = 2$
$u_1(NT, T) = 0.75$	$u_2(T, NT) = 6$
$u_1(NT, NT) = 2.4$	$u_2(NT, NT) = 5$

The payoffs are in form of estimated percent change in GDP. In 2024, the U.S. GDP grew by 2.4 percent (BEA, 2025) and the Chinese GDP grew by 5 percent (Barcelona et al, 2025), giving our baseline payoffs for (*No Tariff*, *No Tariff*). The other payoffs are based on the Trump Administration’s perspective on the effects of levying tariffs¹. The Administration has the perspective that the tariffs will bring growth and benefit the United States overall, while harming foreign countries enough to bring them to the negotiating table. The payoffs reflect this and assume that China knows of these beliefs and plays accordingly. If both countries choose *Tariff*, the perspective of the Administration is that while the U.S. will be harmed by Chinese tariffs, China will be also taking a loss. If the U.S. chooses *Tariff* and

¹It should be noted that several independent agencies, including the [The Yale Budget Lab](#) and the [Congressional Budget Office](#), predict a loss in GDP if any tariffs are implemented, regardless of retaliation. Further, several other effects on employment, consumption, imports, exports, prices, and welfare should be considered for a fuller picture of the effects of a trade war.

China chooses *No Tariff*, the view of the Administration is that it will bring the largest benefit to the U.S. and vice versa. Written in a payoff matrix, the baseline game is:

		China	
		<i>T</i>	<i>NT</i>
United States	<i>T</i>	1, 2	3.4, 1.5
	<i>NT</i>	0.75, 6	2.4, 5

Figure 1: The Baseline Game

Given China chooses either *Tariff* or *No Tariff*, the best response for the United States is to choose *Tariff*. The same is true for China given that the United States chooses either strategy. Therefore, the strictly dominant strategy for both players is *Tariff*, and the strictly dominated strategy is *No Tariff*. The Nash Equilibrium of the game is (*Tariff*, *Tariff*).

III Extensions of The Baseline Game

a. Indefinite Prisoner's Dilemma

With the exception of 2018, the United States has not levied any large scale tariffs since the Great Depression. There have been many trade disputes since then, so why hasn't the United States implemented tariffs until now? What changed to make the equilibrium go from (*No Tariff*, *No Tariff*) to (*Tariff*, *Tariff*)? One probable reason is that the discount factor for the United States has fallen below the level needed to sustain cooperation. Consider an indefinitely repeated prisoner's dilemma; the strategy played by both players resembles the grim trigger strategy. When one country implements tariffs, the other will as well, and the tariffs remain in place for an indefinite time until a deal is reached (maybe a new administration is elected whose discount rate is high enough to sustain cooperation and make a deal). Given our payoffs, we can calculate what discount rates would be needed to sustain cooperation or tariff. Formally, let the discount factor δ be given by the equation

$$\delta = \frac{1}{(1+r)^n} \quad (1)$$

Where $r > 0$ is the interest rate, and therefore $\delta < 1$. Then, the discounted present value of the payoffs P is

$$P \sum_{n=1}^{\infty} \frac{1}{(1+r)^n} = P \sum_{n=1}^{\infty} \delta^n = P + P\delta + P\delta^2 + P\delta^3 + \dots + P\delta^n$$

To find what this series converges to, consider the partial sum δ_m of our series (P is a

constant and is left out)

$$\delta_m = \delta + \delta^2 + \delta^3 + \dots + \delta^m$$

Multiplying both sides by $(1 - \delta)$

$$(1 - \delta)\delta_m = \delta + \delta^2 + \dots + \delta^{m-1} + \delta^m - \delta^2 - \delta^3 - \dots - \delta^m - \delta^{m-1} = \delta - \delta^{m-1}$$

Dividing both sides by $(1 - \delta)$ and taking the limit²:

$$\delta_m = \frac{\delta - \delta^{m+1}}{1 - \delta} \implies \lim_{m \rightarrow \infty} \frac{\delta - \delta^{m+1}}{1 - \delta} = \frac{\delta}{1 - \delta}$$

Bringing back P gives us the discounted present value of P received starting next period and each period after that

$$\frac{\delta}{1 - \delta} \cdot P \tag{2}$$

Now we are ready to calculate the discount factors for our game. For the United States, the payoff for *Tariff* is

$$3.4 + \frac{\delta}{1 - \delta} \cdot 1$$

The payoff for *No Tariff* is

$$2.4 + \frac{\delta}{1 - \delta} \cdot 2.4$$

Tariff is favorable to *No Tariff* if

$$3.4 + \frac{\delta}{1 - \delta} \cdot 1 \geq 2.4 + \frac{\delta}{1 - \delta} \cdot 2.4 \implies \delta \leq \frac{5}{12}$$

A discount factor less than or equal to $\frac{5}{12}$ reflects that the U.S. is less patient and prioritizes the short term, therefore choosing *Tariff*. Discount factors can also be used to explain why some adversaries, like the European Union, have not retaliated yet. In April of 2025, the European Commission President stated, “We took note of the announcement by President Trump. We want to give negotiations a chance...If negotiations are not satisfactory, our countermeasures will kick in” ([Reid, 2025](#)). Adjusting for the E.U’s GDP growth of 1 percent in 2024 ([Eurostat, 2025](#)), *No Tariff* is favorable to *Tariff* if

$$1 + \frac{\delta}{1 - \delta} \cdot 1 \geq 2 + \frac{\delta}{1 - \delta} \cdot 0.5 \implies \delta \geq \frac{2}{3}$$

²If the limit of a partial sum exists, then our series is convergent. The proof is based off [Games of Strategy](#) by Dixit, A.K., Skeath, S., and Reiley, D and notes from ECON 52295

A discount factor greater than or equal to $\frac{2}{3}$ reflects that the E.U. is more patient and prioritizes the long term, therefore choosing not to retaliate and negotiate instead (at least until they start to become less patient during trade talks).

b. A Game with Alternating Offers

The baseline game predicts that both the United States and China should levy tariffs on each other. This, however, did not last for some opponents of the United States. On November 14, 2025, the Administration lifted global tariffs on some agricultural products like coffee, tea, beef, and bananas (Bown, 2025). Also, on November 20, 2025, the Trump Administration suddenly lifted the extra tariffs on several products from Brazil, including nuts, coffee, cocoa, and beef (Bown, 2025). The Administration cited affordability concerns as the reason for the moves. What did the U.S. gain from these deals?

Consider a game with alternating offers, played by the U.S. and an Adversary, over an indefinite time frame. The gains from trade, M , are in the form of the affordability benefits to consumers from lifting tariffs³ Let the U.S. be PY and the Adversary be PX . In the first round, the Adversary makes an offer, x , and the U.S. can either *Accept* (A) the offer, *Walk Away* (W), or make a counteroffer. If the U.S. accepts, the game ends and the Adversary receives $(M - x_t)$ and the U.S. receives x_t . If the U.S. walks away both players receive a negative payoff (continued harm from the tariffs). This general form continues throughout the game. The Adversary makes offers in the odd numbered rounds, and the U.S. makes offers in the even numbered rounds. The normal form of the game is

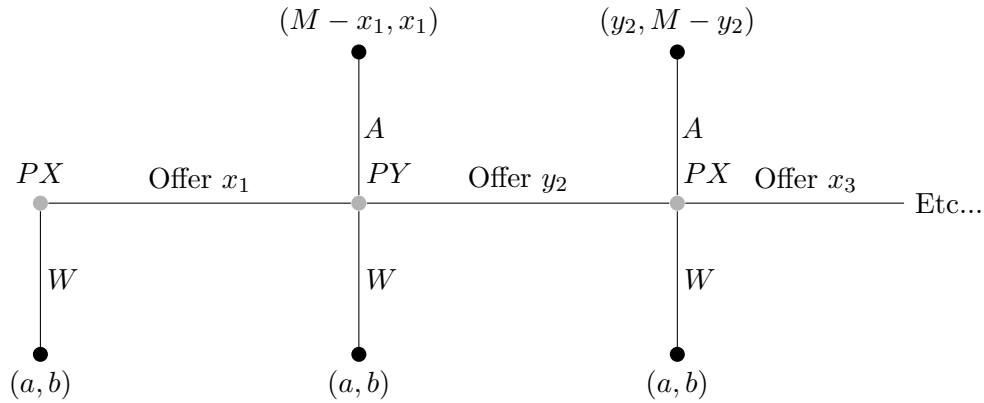


Figure 2: A Game With Alternating Offers

³Lower prices can lead to increased consumption and therefore a boost to growth. Tariffs also result in less variety of goods available to consumers (Amiti et al., 2019).

Where $a, b < 0$. We know both countries discount future payoffs, and the United States now has a lower discount rate than before. Let the discount rate of the Adversary be δ_X and the discount rate of the U.S. be δ_Y . To find the equilibrium, we determine the Adversary's offer such that the U.S. would accept it in the current period, knowing that a counteroffer made in the next round would also be accepted. Given an offer x_t from the Adversary, if the U.S. rejects they receive $\delta_Y(1 - y_t)$ or if the U.S. accepts, they get x_t , and vice versa. The smallest offer that the U.S. accepts is one that makes them indifferent to A or W

$$x_t = \delta_Y(1 - y_t) \quad (3)$$

The smallest offer that the Adversary accepts is one that makes them indifferent to A or W ,

$$y_t = \delta_X(1 - x_t) \quad (4)$$

Solving for the equilibrium offers,

$$x^* = \frac{\delta_Y - \delta_X \delta_Y}{1 - \delta_X \delta_Y} \quad (5)$$

$$y^* = \frac{\delta_X - \delta_X \delta_Y}{1 - \delta_X \delta_Y} \quad (6)$$

In [Section IIIa](#), I found the discount rate of the U.S. to be at most $\frac{5}{12}$. Assuming the Adversary has a higher discount rate, like that of the E.U, we can calculate the potential shares of the gains from trade for each country. Let $\delta_X = \frac{2}{3}$ be held constant.

U.S Discount Rate (δ_Y)	U.S Share (x^*)	Adversary's Share ($1 - x^*$)
0.10	0.0357	0.9643
0.2	0.0769	0.9231
0.30	0.1250	0.8750
0.4167	0.1923	0.8077
0.50	0.2500	0.7500
0.70	0.4375	0.5625
0.90	0.7500	0.2500

Table 1: The Share of The Gains from Trade, Holding δ_X Constant

In equilibrium, the lower discount rate of the U.S. results in a lower share of the gains from trade; being more patient results in higher shares. At the discount rate calculated in [Section IIIa](#) (highlighted in blue), the U.S. gets about a 19 percent share. One reason the

Trump Administration implemented the tariffs was to force other nations to the negotiating table, so the U.S. could gain more favorable trade terms. However, the new lower discount rate resulted in the U.S. gaining less of the shares of gains from trade than might of been anticipated. In the game, the Adversary also has a first mover advantage, which results in a higher share for them. The rescission of tariffs and low discount rate by the U.S. results in a weaker position for other trade goals, especially against smaller nations who may not be able to sustain a trade war as long. This game can also be used to explain why some Adversaries of the United States, like China, have reached deals to bring down large tariff rates.

IV Discussion

The 2025 Trade War is complex and fast moving, with different adversaries and various stages of the war being modeled by different tools from game theory; the most basic of which is a prisoner's dilemma. I model payoffs in terms of the change in 2024 GDP based on Trump Administration's perspective on the effects of tariffs. The Administration believes that levying tariffs will be beneficial for the United States and harmful for China, though less beneficial if China retaliates. For both the United States and China, the strictly dominant strategy is *Tariff*, and the Nash Equilibrium is (*Tariff*, *Tariff*). This equilibrium, while beneficial in the view of the Administration, is not optimal for the United States, and is also a Bayes-Nash Equilibrium. The decision by the U.S. to tariff may be a potential agency problem, especially if tariffs are viewed as a negotiation tool, and not one which brings economic benefit.

The Nash Equilibrium of the Baseline Game does not hold up in all cases, however; it was not what we observed for the greater half of a century. This deviation may be due in part to the discount factor of the United States falling below the threshold needed to sustain free trade. Extending the one-shot Baseline Game to an indefinite repeated prisoner's dilemma where both countries play grim trigger, I find the discount rate of the United States must be less than or equal to $\frac{5}{12}$ in order to levy tariffs. One reason why some countries, like the E.U, have not retaliated is because their discount rate may be high enough to sustain cooperation; they may believe they can reach a deal before they need to retaliate. Interestingly, the discount rate of the United States may be fluctuating between recent administrations, with the Biden Administration undoing most of the tariffs levied by Trump in 2018-19. The different patience levels may also be influenced by international relations and politics, making the U.S. less patient with China than other countries.

The U.S. has made some trade deals, as we see with the rescission of some tariffs on agricultural goods. I model this observation with an indefinite game with alternating offers between the U.S. and an Adversary. The gains from trade are in the form of affordability benefits to consumers. The Adversary makes an offer, and the U.S. may either accept the offer, walk away from the offer, or make a counteroffer. After calculating the equilibrium offers and shares, I find that the United States could increase its share of the gains from trade by having a higher discount rate. At the discount rates calculated in [Section IIIa](#), the U.S. only gets about a 19 percent share of the gains from trade. Being less patient results in a weaker position for the United States. The new lower U.S. discount rate may result in less favorable outcomes in future deals, especially if a given adversary has a higher discount rate than the United States.

The complex and ever-changing nature of the trade war allows for a variety of game theory tools to be applied but also complicates them. The internal inconsistency of the two Trump Administrations makes it difficult to definitively define their beliefs. So, this paper relies on stated beliefs and assumes they remain consistent until stated otherwise. There are also countless political factors that may have significant influence on the beliefs of both the U.S. and foreign leadership. The payoffs, and therefore discount rates, are proxies for the beliefs of the Trump Administration and a given adversary. Further exploration is needed to explain why the E.U. retaliated quickly during the 2018 Trade War but not so during the 2025 Trade War. Further, one could model a game with incomplete information where the E.U. determines how they would respond given the result of the 2024 election. The game with alternating offers can also be used to model other trade deals the U.S. has made, especially the one with China to bring down the unprecedented tariff rates we saw in April. As the 2025 Trade War progresses, game theory can provide insights into the strategies of international trade.

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